TECHNICAL ARCHITECTURE & MODEL DOCUMENTATION

DeepSentinel AI

Detecting manipulated images, audio and video, end to end: the technology stack, the models and why each was chosen, the system architecture, the API, and what has and has not been validated.

FastAPI · SQLAlchemy · Alembic | Celery + Redis | PostgreSQL 16 | PyTorch · torchvision · transformers | React 18 · Vite 5 · Tailwind 3 | Docker Compose · nginx

> **Automated assessment only. Not forensic or legal evidence.** Every metric in this document is copied from a file the repository's own scripts wrote; Appendix A lists the file for every number. Signals that have not been validated on labelled data are marked **UNVALIDATED**.

Repository
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How to read this document

Each component has a *Why* box explaining the design decision. Status badges show how much trust a signal has earned: TRAINED & EVALUATED a model trained in this repo with a held-out evaluation file · THIRD-PARTY PRETRAINED public weights, chosen by a measurement in this repo · INFORMATIONAL computed and shown, but not used in the verdict · UNVALIDATED HEURISTIC hand-set rule, no labelled evaluation · IN PROGRESS not on this branch.

1 Executive summary

DeepSentinel AI is a web platform that takes an uploaded image, audio file or video, runs it through a detection pipeline for that media type, and returns a **probability that the media is manipulated or AI-generated**, a three-way verdict (`AUTHENTIC` , `MANIPULATED` , `INCONCLUSIVE`), visual evidence (heatmaps, spectrograms, timelines), and a downloadable PDF report with a SHA-256 hash.

Image

Two public scene-level AI-image detectors fused in log-odds space (0.75 / 0.25) make the decision. A face-crop EfficientNet-B4 is shown but kept out of the verdict. A forensic suite (ELA, CFA, metadata, C2PA) adds context.

Audio

A Light CNN (LCNN) over LFCC features, trained in this repo on ASVspoof 2019 LA, is the only input to the audio verdict. It scores each 4-second window to build the timeline.

Video

Samples 1 frame/s (max 32), crops the largest face with MTCNN, scores frames, runs the scene ensemble on the worst frame, then 11 stages with hand-set fusion. A trained face-swap model is IN PROGRESS.

Headline evaluated result

4.78%	0.9865	5.26 ± 0.65%
AUDIO EER, SERVED SEED	AUDIO AUC, SERVED SEED	AUDIO EER, 3 SEEDS
ASVspoof 2019 LA eval · 71,237 utts	bona fide vs spoof	mean ± SD

Source: results/audio_eval.json and paper_results/audio_asvspoof2019la/metrics.json (Appendix A).

What this evaluation does and does not cover

The audio model is the only detector here with a held-out evaluation file from a public benchmark. That benchmark is studio-quality English read speech. The model has **not** been evaluated on phone calls, real-world recordings or other languages (`LCNN_SCOPE_NOTE` , audio_pipeline.py). The image ensemble was chosen on a 560-image measurement that is recorded only in a source-code comment. The video verdict uses hand-set fusion weights.

Design principles that run through the codebase

- **Measured, not assumed.** Detectors are kept, reweighted or dropped based on measurements. Example: `umm-maybe/AI-image-detector` scored AUC 0.469 and was removed.
- **Refuse rather than guess.** An uncertainty band around the threshold returns `INCONCLUSIVE` . An unflagged screenshot is INCONCLUSIVE, never AUTHENTIC.
- **Weights must match their calibration.** The audio checkpoint is loaded only if its SHA-256 matches the calibration file. A missing image checkpoint is flagged `untrained-backbone` everywhere.
- **Heuristics carry zero weight until proven.** Audio heuristics get weight only if their bootstrap 95% AUC lower bound on dev is above 0.6. None passed.

2 Technology stack and why

Versions are the lower bounds in `backend/requirements*.txt`, `frontend/package.json` and the Dockerfiles. Exact tested versions are pinned in `backend/requirements.lock`.

Backend and ML

Layer	Technology	Why this choice
Language	Python 3.11 python:3.11-slim image	The ML ecosystem (PyTorch, librosa, OpenCV, transformers) is Python-first. Keeping the API and inference in one language lets the API import the pipelines directly in eager mode.
Web framework	FastAPI ≥ 0.115 + Uvicorn	Async I/O for uploads and WebSockets, Pydantic validation of every request and response, and OpenAPI docs generated at `/docs`.
Validation / settings	Pydantic 2, pydantic-settings	One typed `Settings` class reads env vars and `.env`. Production refuses to boot with the placeholder JWT secret (`InsecureConfigurationError`).
ORM / migrations	SQLAlchemy 2 + Alembic	The same models run on SQLite (local dev) and PostgreSQL (production). Alembic owns the production schema, so `create_all` never alters a live database.
Database	PostgreSQL 16 (prod), SQLite (dev)	Postgres handles concurrent API and worker writes. SQLite means no setup for local development and tests.
Task queue	Celery ≥ 5.4 + Redis 7	Inference takes seconds to minutes and needs a GPU. A queue keeps HTTP requests fast. Workers run with `--concurrency=1` so only one model copy sits in GPU memory. `beat` schedules the hourly retention purge.
Auth	python-jose (JWT), passlib, httpOnly cookie	A stateless JWT in an httpOnly `SameSite=Lax` cookie can't be read by page scripts. A Bearer header is still accepted for API clients.
Deep learning	PyTorch ≥ 2.2, torchvision ≥ 0.17	torchvision provides ImageNet-pretrained EfficientNet. The LCNN and Xception are defined in-repo (`models_arch.py`), so there's no dependency on the unmaintained `pretrainedmodels` package.
Pretrained hub models	Hugging Face transformers ≥ 4.30	Loads the two scene-level AI-image detectors through `pipeline("image-classification")`. Cached in a named volume (`HF_HOME`), so weights download once.
Face detection	facenet-pytorch (MTCNN)	Deepfake artefacts are concentrated in the face, and FF++-style models expect face crops. If MTCNN can't load, the pipeline falls back to the full frame and says so in the notes.
Signal / media	librosa, scipy, soundfile, OpenCV, imageio-ffmpeg, imagehash, Pillow, pillow-heif	Audio decoding and features, video frame sampling and optical flow, perceptual hashes, and HEIC photos from phones.
Reports	ReportLab + Matplotlib	Generates PDFs server-side without a headless browser. Matplotlib draws the timeline and score charts embedded in the report.

Frontend

Layer	Technology	Why this choice
UI library	React 18.3 + React Router 6	The result pages are component-heavy (tabs, per-media views, pipeline visualisers). Routes: upload, result, report, history, privacy.
Build tool	Vite 5.4	Fast dev server with an `/api` proxy (including WebSockets) to port 8000, so the browser sees one origin in development. Recharts gets its own chunk for cache stability.
Styling	Tailwind CSS 3.4 + PostCSS	Design tokens (status colours, surfaces) shared by all forensic views.

Layer	Technology	Why this choice
Charts / motion	Recharts 2, Framer Motion, GSAP	Per-window and per-frame score timelines, and UI transitions.
Testing / lint	Vitest 2 + Testing Library + jsdom, ESLint 8	Component tests render real API responses from fixture JSON exported from actual jobs.
Serving	nginx 1.27-alpine	Serves the static build and reverse-proxies `/api`. It has a fixed IP (172.28.0.10), so the API trusts `X-Real-IP` only from it.

Infrastructure

Concern	Technology	Why this choice
Orchestration	Docker Compose	Six services (db, redis, api, worker, beat, frontend) come up with one command. Named volumes: `db_data`, `redis_data`, `media`, `hf_cache`.
TLS / edge	Caddy or Cloudflare Tunnel	`deploy/Caddyfile` and `deploy/cloudflare-tunnel.yml` cover a public VM and a lab machine with no public IP.
Ops	pg_dump backups, uptime check	`deploy/backup.*` and `deploy/uptime_check.*` probe `/api/health`.
CI	GitHub Actions	`.github/workflows/ci.yml` builds the backend image, runs pytest inside it, then runs `npm ci`, tests and a build for the frontend.

3 System architecture

The system has two modes, chosen by whether `CELERY_BROKER_URL` is set. In **queued mode** (Docker Compose) the API never loads a model, and only the worker holds models in memory. In **eager mode** (local development, as currently running) the API runs the pipeline in-process and warms up the models at startup.

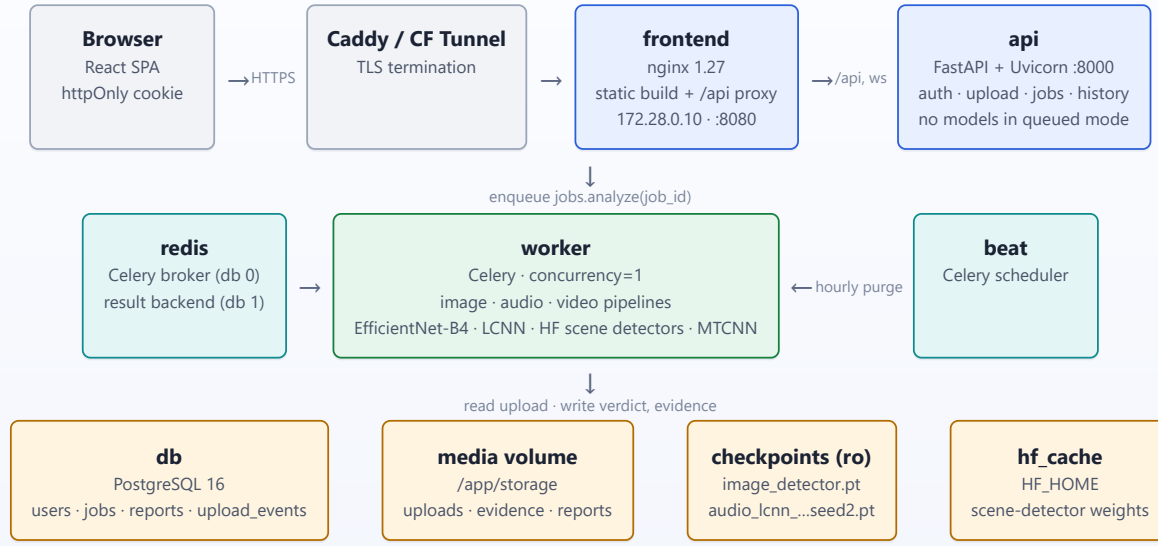


Figure 1: Container topology in queued mode (docker-compose.yml). The API and worker share the media volume. Checkpoints and paper/config are mounted read-only.

Backend module map

Path	Responsibility
`app/main.py`	App factory, lifespan (directories, DB init, model warm-up only when no broker is set), CORS, security headers, request-ID middleware, early 413 on oversized uploads, error handlers that never leak stack traces.
`app/routers/`	`health`, `auth`, `uploads`, `jobs`, `history`, all mounted under `/api`.
`app/worker/`	`celery_app.py`, `tasks.py`: `jobs.analyze` (1 retry) and `maintenance.purge_expired_guest_jobs`.
`app/ml/pipeline.py`	Dispatches by media type to the three live pipelines.
`app/ml/image_pipeline.py`	Recapture cropping, MTCNN faces, scene ensemble, Grad-CAM, forensic suite, final probability.
`app/ml/audio_pipeline.py`, `app/ml/audio/`	LCNN verdict, per-window timeline, context signals (weight 0).
`app/ml/video/__init__.py`	11-stage video orchestrator and hand-set fusion (`video/fusion.py`).
`app/ml/registry.py`	Loads each model once per process (thread-safe cache), records `weights_status`, enforces the audio checkpoint hash.
`app/report/`	ReportLab PDF generator and Matplotlib charts.
`app/signing.py`, `app/security.py`, `app/deps.py`	HMAC-signed short-lived URLs, password hashing / JWT, request dependencies (current user, guest access).
`app/retention.py`, `app/storage.py`	Guest-job expiry and file deletion. Upload storage and SHA-256 hashing.
`app/ml_v1/`	Legacy copy. **Never edited, never imported by live pipelines.**

4 End-to-end request lifecycle

This is what happens between a user dropping a file on the Upload page and downloading the PDF.

1. **Limits check.** The SPA calls `GET /api/upload/limits` for allowed extensions, the size limit (`MAX_UPLOAD_MB` , default 100), duration limits (audio 600 s, video 300 s), enabled media types and guest retention hours.
2. **Upload.** `POST /api/upload` (multipart). Middleware rejects an over-size `Content-Length` with 413 before reading the body. The router checks the media type is enabled (else 403), checks duration (else 400), applies per-user/IP rate limits (`upload_events`), stores the file, computes SHA-256, and creates a `jobs` row with status `queued` .
3. **202 Accepted.** The response contains `id` , `case_reference` , `sha256` , `file_size_bytes` , `uploaded_at` .
4. **Enqueue.** `enqueue_analysis(job_id)` sends `jobs.analyze` to Redis. In eager mode it runs in-process.
5. **Live progress.** The SPA calls `POST /api/jobs/{id}/links` for signed `ws_url` / `report_url` (5-minute TTL), then opens `/api/jobs/{id}/ws` . It falls back to polling `GET /api/jobs/{id}/status` .
6. **Inference.** The worker sets `processing` and calls `analyze(media_type, path, EVIDENCE_DIR, job_id)` . The pipeline returns an `AnalysisResult` : probability, verdict (or a verdict override for audio), confidence, model name/version, `weights_status` and an evidence dictionary. Evidence images are written as `EVIDENCE_DIR/<job_id>_*.png` .
7. **Persist.** The verdict, probability, model info, JSON-safe evidence and `processing_ms` are written to the job. Status becomes `done` , or `failed` with a user-facing message (corrupt media, missing detector, missing file).
8. **Result.** `GET /api/jobs/{id}/result` returns `JobResultOut` , including `experimental` , `experimental_reason` and the fixed `disclaimer` . Evidence images come from `GET /api/jobs/{id}/evidence/{kind}` .
9. **Report.** `POST /api/jobs/{id}/report` builds the PDF, stores its SHA-256 in `reports` , and returns `report_reference` and `download_url` . `GET /api/jobs/{id}/report` streams it with an `X-Report-SHA256` header. `GET /api/reports/{ref}/verify` checks a report reference.
10. **Deletion / retention.** `DELETE /api/history/{id}` removes the upload, every evidence file, the report PDFs and the database rows. Guest jobs are purged automatically after `RETENTION_HOURS` (72).



Figure 2: Job status machine (JobStatus enum in app/models.py). `jobs.analyze` retries once with a 10-second delay on unexpected errors. User errors are not retried.

5 Model inventory

Every component that produces a score, with its status. The **In verdict?** column is the most important: it shows which numbers can actually change the result a user sees.

Component	Media	Status	In verdict?	Architecture / source
haywoodsloan/ai-image-detector-deploy	Image, video keyframe	THIRD-PARTY PRETRAINED	Yes (weight 0.75)	SwinV2 image classifier from the Hugging Face Hub.
Organika/sdxl-detector	Image, video keyframe	THIRD-PARTY PRETRAINED	Yes (weight 0.25)	Swin classifier, Stable Diffusion / SDXL specialist.
EfficientNet-B4 face model checkpoints/image_detector.pt	Image faces, video frames	INFORMATIONAL	No (`FACE_MODEL_IN_VERDICT=false`)	torchvision EfficientNet-B4, ImageNet init, 1-logit head, 224 px input. Trained in-repo on face crops.
LCNN (LFCC), ASVspoof 2019 LA audio_lcnn_asvspoof_lfcc_seed2.pt	Audio, video soundtrack	TRAINED & EVALUATED	Yes (sole input)	Light CNN with Max-Feature-Map activations over 60×401 LFCC windows. Trained in-repo, 3 seeds, served seed picked on dev EER.
MTCNN	Image, video	THIRD-PARTY PRETRAINED	Indirectly (crop only)	facenet-pytorch face detector. 25% margin, minimum box 48 px, largest face first.
Hand-set provenance floors	Image	UNVALIDATED HEURISTIC	Yes (raises the floor)	C2PA AI disclosure → ≥ 0.97; generator metadata → ≥ 0.93; multi-signal consensus → ≥ 0.88; corroboration → ≥ 0.82.
Image forensic suite (ELA, CFA, stego, hashes, metadata)	Image	UNVALIDATED HEURISTIC	Via floors above	Classical signal processing. The face-seam check is reported only (fired on 30% of real portraits).
Video stages 5–9, 11 (jitter, rPPG, 4D tensor, lip kinematics, optical flow, generator profile)	Video	UNVALIDATED HEURISTIC	Yes (hand-set fusion)	Hand-set thresholds and weights in `video/fusion.py` . Generator "profiles" named after Sora/Veo/Kling/Runway/Luma have never been tested on labelled video.
Audio glottal, ENF, splice, container, watermark	Audio	UNVALIDATED HEURISTIC	No (weight 0)	Dev AUCs 0.43–0.57, all below the 0.6 inclusion bar.
Face-swap video model	Video	IN PROGRESS	Not on this branch	Being wired on `ml/faceswap-wire` . Dataset from FF++ (c23) and Celeb-DF v2. No evaluation file exists on this branch.
Retired: `audio_detector.pt`	-	NEVER LOADED	No	Trained on NumPy-generated tones, not speech. Its "AUC 1.0" was measured on tones.

6 Image pipeline

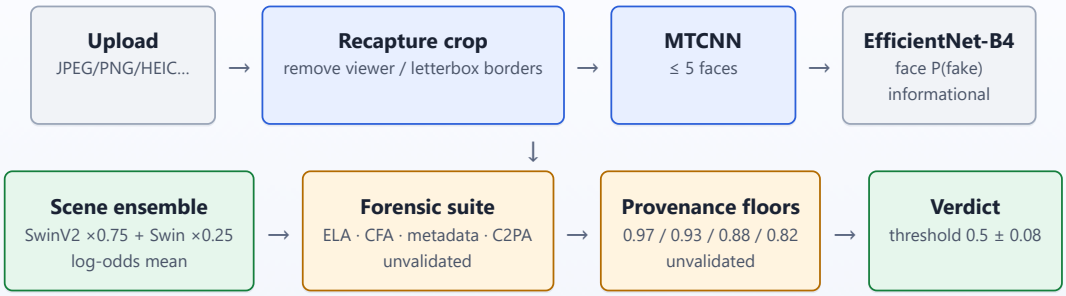


Figure 3: Image flow (image_pipeline.py). Grad-CAM is computed on the highest-scoring face crop.

6.1 Scene-detector ensemble THIRD-PARTY PRETRAINED

For each detector, *P(AI) = 1 − P(real class)*, where the real class is found in that model's own label map ("real", "human", "hum", "authentic"). This stops a model with unusual label names from being silently mis-scored. The fused score is `sigmoid(Σ wᵢ·logit(pᵢ) / Σ wᵢ)`.

Why these two models and this fusion? They were chosen by measurement, not by model card. A comment in `image_pipeline.py` records a test on 560 held-out images: 200 real COCO/Flickr photos and 360 from SD 2.1, SDXL, SD3, DALL-E 3, Midjourney v6, Flux.1-dev, and Gemini "Nano Banana" / Nano Banana Pro. Each was scored as the original file, as a normalised JPEG and as a simulated screenshot. Log-odds averaging was chosen over max-of-experts so that one over-eager model can't flag a real photo on its own.

Configuration (from source comment)	AUC	Real photos scored ≥ 0.5
haywoodsloan alone	0.973	3.5%
0.75 / 0.25 fusion with Organika (served)	**0.984** (0.980 on held-out halves)	3.5%
Organika alone	0.836	23%
umm-maybe/AI-image-detector	0.469 (below chance, removed)	-
Max-of-experts (old rule)	-	26–36%

Provenance caveat

These figures appear only in a source-code comment. No results file for them exists in `results/` or `paper_results/` on this branch. The script `ml/evaluation/eval_image_detectors.py` exists. Re-running it and committing its output file would make these numbers citable.

6.2 EfficientNet-B4 face model INFORMATIONAL

A torchvision EfficientNet-B4 (ImageNet init) with its classifier swapped for a single logit, trained at 224 px. It runs in float32, because float16 autocast overflowed to NaN on some inputs. Its per-face score and Grad-CAM heatmap are shown, but `face_model_in_verdict=False` keeps it out of the verdict.

Why is it excluded from the verdict? The comment in `config.py` says it was trained on StyleGAN faces only. On real vs modern-AI photos with faces it flagged 21% of real faces and only 15% of AI faces, so it can't tell them apart on current generators. It stays out until a retrained model is validated.

Checkpoint summary field	Value
Validation accuracy / AUC / EER (threshold 0.5)	0.99977 / 0.999999 / 0.00015
Confusion matrix (tn / fp / fn / tp)	81,968 / 32 / 5 / 81,995
Train / validation size; epochs	936,009 / 164,000; 1 epoch, AMP
Recorded model_version	`image-detector-v2.1.0-b4-expanded`

Do not read these as real-world accuracy

These are **in-distribution validation** numbers from `checkpoints/image_detector.summary.json`. The validation split comes from the same sources as training. The 21% / 15% result above shows how far they are from out-of-distribution performance. Also, `/api/health` reports the *configured* version `image-detector-v3.0.0-b4-800k`, while the checkpoint summary records v2.1.0. The authoritative version for any analysis is the one stored on its job.

6.3 Screenshots and forensic suite UNVALIDATED

- **Recapture handling:** uniform viewer bars are cropped before any pixel analysis. A screenshot that the detectors don't flag is reported `INCONCLUSIVE`, because recapture destroys the evidence an AUTHENTIC verdict needs.
- **Camera-trace absence** (no CFA trace) only corroborates a detector that has already fired. It was missing in all 196 real web JPEGs measured (source comment), so on its own it means nothing.
- **Declared provenance** (C2PA `trainedAlgorithmicMedia`, generator parameters in metadata) raises the floor. The floor values are hand-set.

7 Audio pipeline

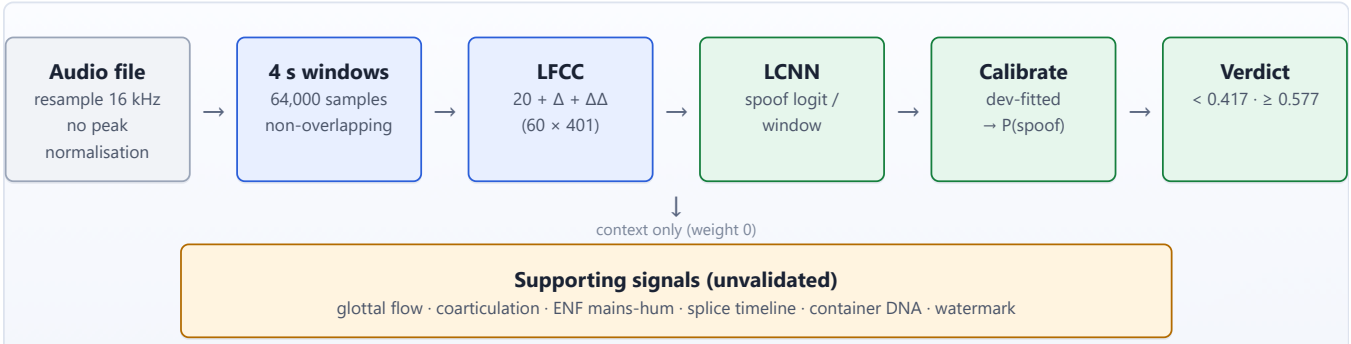


Figure 4: Audio flow. Features are computed by asvspoof_frontend.py, the same code used in training.

7.1 Front end: LFCC TRAINED & EVALUATED

Each window: 20 ms Hamming (320 samples) zero-padded to a 512-point FFT, 10 ms hop, centred frames, 20 linear triangular filters from 0 to 8 kHz, natural log with a 1e-10 floor, orthonormal DCT-II keeping c0–c19, plus deltas and double deltas over 3 frames. That gives a **60 × 401** matrix. Features are rounded through float16 exactly as during training.

> **Why LFCC?** The front end was chosen on the dev set by `ml/evaluation/select_asvspoof_frontend.py`; a log-Mel checkpoint also exists for comparison. Linear filters keep high-frequency resolution where vocoder artefacts tend to sit. Serving imports the training feature code, so the deployed model sees exactly the features it was evaluated on. That's why the waveform is deliberately *not* peak-normalised: normalising shifts c0.

7.2 Network: Light CNN (LCNN)

Nine convolution blocks with **Max-Feature-Map** activations (split channels in half, take the element-wise max), BatchNorm and four 2×2 max-pools. Then global average pooling and a head of Dropout 0.3 → Linear 32→64 → BN → ReLU → Linear 64→1. The checkpoint is about 600 KB.

> **Why LCNN?** It's the architecture used by the ASVspoof anti-spoofing baselines, so results can be compared with published systems. It's small enough to run on CPU. Max-Feature-Map acts as a learned feature selector, which suits narrow spectral artefacts.

7.3 Serving safeguards

- The checkpoint named in `paper/config/audio_calibration.json` is loaded **only if its SHA-256 matches** (`7ca31fd0…`). Any other weights make the detector unavailable, rather than producing probabilities nobody measured.
- The verdict thresholds are frozen on dev in `frozen_audio_config.json`: decision threshold 0.497 from dev, band ±0.08. That gives AUTHENTIC < 0.417, MANIPULATED ≥ 0.577, INCONCLUSIVE in between.
- `AUDIO_MODEL_SOURCE=legacy` switches back to hand-set rules and marks every result `experimental: true`.

7.4 Evaluation: ASVspoof 2019 LA eval

4.78%	0.1025	8.09%
EER (SERVED SEED 2)	MIN T-DCF (SEED 2)	PUBLISHED B02 LFCC-GMM
7,355 bona fide · 63,882 spoof	3-seed mean 0.1104 ± 0.0142	B01 CQCC-GMM: 9.57%

Attack	A07	A08	A09	A10	A11	A12	A13	A14	A15	A16	A17	A18	A19
EER %, 3-seed mean	0.32	1.74	0.10	0.78	0.41	0.30	0.39	2.44	2.74	0.55	**26.90**	2.06	3.53

Known weak spot: attack A17

EER is 26.90% (mean of 3 seeds), far above every other attack. Any spoofing method similar to A17 will often get through. The model is also untested outside ASVspoof's studio-quality English read speech.

7.5 Why the heuristics get zero weight

The inclusion rule in `frozen_audio_config.json`: a signal gets weight above 0 only if its bootstrap 95% AUC lower bound on the balanced dev subsample (1,000 bona fide + 1,000 spoof) is above 0.6. On that subsample the LCNN scored AUC 1.000. Every heuristic sat near chance:

Signal	Dev AUC [95% CI]	Signal	Dev AUC [95% CI]
Glottal anomaly	0.552 [0.527, 0.578]	Splice worst segment	0.499 [0.498, 0.500]
Coarticulation anomaly	0.518 [0.494, 0.543]	Splice F1 spikes	0.428 [0.404, 0.453]
Environment / mains-hum	0.504 [0.499, 0.509]	Legacy final probability	0.556 [0.543, 0.569]

The legacy rule-based verdict labelled 99.1% of bona fide and 87.0% of spoof clips INCONCLUSIVE, which is why it was replaced.

8 Video pipeline

The served video detector, as `/api/health` reports it, is "*frame-aggregation over the image model*", version `video-frame-agg-v1.0.0`. The orchestrator in `app/ml/video/__init__.py` runs 11 stages and records the timing and status of each in `pipeline_modules`.

#	Stage	Status	What it does
1	Container & bitstream	UNVALIDATED	Container format, atom hierarchy, screen-recording detection, compression fingerprint.
2	Frame sampling & shots	PRE-PROCESSING	`VIDEO_SAMPLE_FPS` =1, `VIDEO_MAX_FRAMES` =32. Shot-boundary detection.
3	Spatial neural + GenAI	PRETRAINED / INFO	MTCNN largest face → EfficientNet-B4 per frame (informational). Scene ensemble on the worst face frame. Frames with no face are excluded, because the face model scores scenery as noise.
4	Image forensic suite on keyframe	UNVALIDATED	CFA periodicity, Fourier slope, ELA, matte voids.
5	Temporal jitter & facial dynamics	UNVALIDATED	Box jitter, landmark jitter, blink rhythm.
6	rPPG	UNVALIDATED	CHROM/POS pulse estimation, 0.75–2.5 Hz band.
7	Spatio-temporal tensor	UNVALIDATED	Space-time patch variance statistics (named after TimeSformer/SlowFast; no such trained model is loaded).
8	Lip kinematics	UNVALIDATED	Mouth aspect ratio trajectory, articulatory jerk.
9	Optical flow	UNVALIDATED	Farneback dense flow, face-vs-background divergence.
10	Audio track & lip-sync	LCNN + UNVALIDATED	Demuxes the soundtrack and runs the audio suite (LCNN). Audio-visual correlation is a heuristic.
11	Generator profile match	UNVALIDATED	Hand-set profiles named after Sora, Veo, Kling, Runway and Luma. **Never tested on labelled video.**

How the current video score is produced

`fuse_video_forensics()` combines these signals with **hand-set weights**. Because the face model is informational, the spatial input is 0 unless `FACE_MODEL_IN_VERDICT` is true, so in practice the scene ensemble and the heuristics drive the score. No labelled-video evaluation of this fused score exists in the repository.

8.1 Face-swap model IN PROGRESS

A trained face-swap detector is being built to replace the hand-set face signal:

- **Data:** face crops from FaceForensics++ (c23) and Celeb-DF v2, 32 frames per video, the same MTCNN setup and margin as `backend/app/ml/faces.py`, official splits by video, and an audit that fails if any video or identity appears in two splits.
- **Scoring contract the UI already supports** (`VideoForensicsView.jsx`): `face_tracks[]`, one entry per tracked face, scored as the **mean of the top 25% of its frame scores**. The video score is the **highest track score**. A calibrated `segment_timeline` is built from the same scores. With no face track, the UI says "*No face found: face-swap check not applicable.*"
- **Status:** the backend wiring lives on `ml/faceswap-wire`, which is not merged into this branch and has no checkpoint or results file here. **No accuracy figure is claimed for it.**

9 Decision logic and verdicts

Image and video (platform default)

`FAKE_THRESHOLD` = 0.50, `UNCERTAIN_BAND` = 0.08

- p < 0.42 → **AUTHENTIC**
- 0.42 ≤ p < 0.58 → **INCONCLUSIVE**
- p ≥ 0.58 → **MANIPULATED**

app/config.py and app/ml/base.py `classify()`. Pipeline-specific overrides apply, e.g. an unflagged screenshot → INCONCLUSIVE, no scene detector loaded → AUTHENTIC impossible.

Audio (frozen on ASVspoof dev)

Threshold 0.497 (from dev), band ±0.08

- p < 0.417 → **AUTHENTIC**
- 0.417 ≤ p < 0.577 → **INCONCLUSIVE**
- p ≥ 0.577 → **MANIPULATED**

paper/config/frozen_audio_config.json, set via `verdict_override` in audio_pipeline.py.

Why a three-way verdict?

Refusing is a feature. Scores near the threshold are where detectors make most of their mistakes. Returning INCONCLUSIVE there, and for inputs whose evidence was destroyed (screenshots, missing detectors), is more honest than forcing a binary answer. "No manipulation detected" never means "proven authentic".

Fields every result carries

Field	Meaning
`fake_probability`	The pipeline's final probability (for audio, the calibrated LCNN probability alone).
`verdict` , `confidence`	Three-way verdict and its confidence.
`model_name` , `model_version` , `weights_status`	Which model produced the score. `untrained-backbone` is shown everywhere if weights were missing.
`experimental` , `experimental_reason`	True when no trained, evaluated model made the verdict. The UI shows an "Experimental" badge.
`disclaimer`	"Automated assessment only. Not forensic or legal evidence…"
`evidence.*`	Media-specific: `frame_scores` , `segment_scores` , `segment_timeline` , `pipeline_modules` , `notes` , `*_file` evidence images.

10 API endpoint reference

Base path `/api` (`API_V1_PREFIX`). Interactive docs at `/docs` , schema at `/openapi.json` . Authentication is the `deepsentinel_session` httpOnly cookie or `Authorization: Bearer` . Guests can upload and read their own jobs. Every error body includes a `request_id` .

Method	Path	Auth	Purpose / response
Health			
GET	`/api/health`	None	`HealthOut` : status, version, queue_enabled, model descriptions. Never loads a model.
Auth (/api/auth)			
POST	`/api/auth/register`	None	201 `Token` . Sets the session cookie. Password 8–128 characters.
POST	`/api/auth/login`	None	`Token` (access_token, expires_in, user). Sets the cookie.
POST	`/api/auth/logout`	Cookie	204. Clears the cookie.
GET	`/api/auth/me`	User	`UserOut` .
Upload			
POST	`/api/upload`	User or guest	202 `JobCreated` . 413 too large · 403 type disabled · 400 too long · 429 rate limit.
GET	`/api/upload/limits`	None	Allowed extensions, size limit, enabled types, max durations, guest retention hours.
Jobs (/api/jobs)			
GET	`/api/jobs/{id}/status`	Owner / guest job	`JobStatusOut` : status, timestamps, error_message.
POST	`/api/jobs/{id}/links`	Owner / guest job	`JobLinksOut` : signed `report_url` , `ws_url` , `expires_at` (TTL 300 s).
WS	`/api/jobs/{id}/ws`	Signed URL / cookie	Live status push until done or failed.
GET	`/api/jobs/{id}/result`	Owner / guest job	`JobResultOut` : verdict, probability, model, evidence, experimental, disclaimer.
GET	`/api/jobs/{id}/evidence/{kind}`	Owner / guest job	Evidence PNG (heatmap, spectrogram, timeline, optical flow, …).
POST	`/api/jobs/{id}/report`	Owner / guest job	201 `ReportOut` : report_reference, sha256, download_url.
GET	`/api/jobs/{id}/report`	Signed URL / cookie	PDF stream with an `X-Report-SHA256` header.
History			
GET	`/api/history`	User	`HistoryPage` (items, total, limit, offset).
DELETE	`/api/history/{id}`	Owner	204. Deletes the upload, all evidence files, report PDFs and rows.
GET	`/api/reports/{ref}/verify`	None	Checks a report reference against the stored hash.

Example: upload and fetch a result

```
curl -X POST http://localhost:8000/api/upload -F "file=@clip.wav"
→ 202 {"id":"9f3c…","case_reference":"…","media_type":"audio","status":"queued","sha256":"…"}

curl http://localhost:8000/api/jobs/9f3c…/result
→ 200 {"status":"done","verdict":"INCONCLUSIVE","fake_probability":0.51,
       "model_name":"LCNN (LFCC), ASVspoof 2019 LA","experimental":false,
       "evidence":{"segment_scores":[{"start_s":0,"end_s":4,"fake_probability":…,"status":"…"}], …},
       "disclaimer":"Automated assessment only. Not forensic or legal evidence. …"}
```

The values in this example are illustrative placeholders, not measured results.

11 Data model

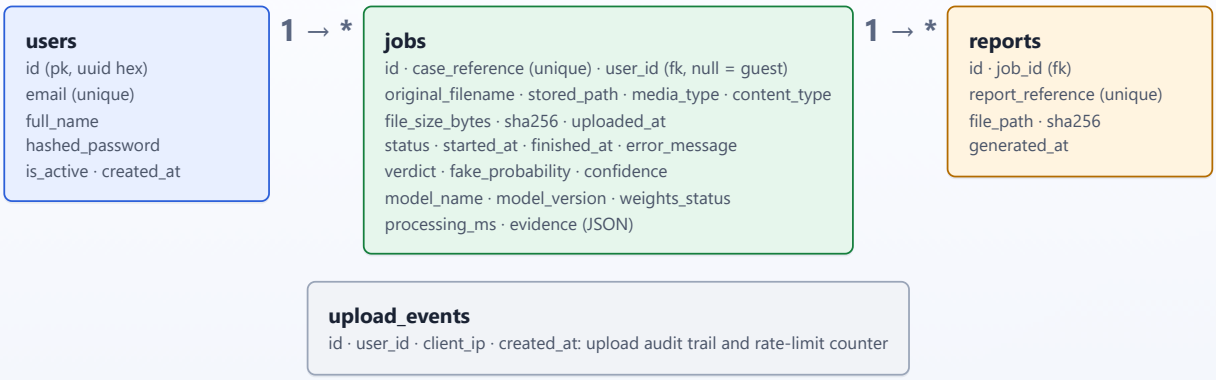


Figure 5: ORM schema (app/models.py). Deleting a user or job cascades to its jobs and reports. A `SafeEnum` type maps legacy enum strings.

Why store evidence as JSON? Each media type produces a different evidence shape (frame scores, window scores, pipeline stages). One JSON column keeps the schema stable while pipelines change. The API contract for each field is documented in `docs/API_CHANGES.md`.

12 Frontend architecture

Pages (src/pages)

Page	Role
Landing	Product overview
Upload	Drag-and-drop, limits from `/upload/limits`, privacy link
Result	Verdict, disclaimer banner, per-media forensics view
Report	PDF generation and signed download
History	Past jobs, delete
Login	Register / sign in (cookie session)
HowItWorks	Method explainer (vendor claims removed)
Privacy	What is stored, 72 h guest retention, deletion, no legal standing

Result views (src/components)

Component	Shows
ImageForensicsView	Scene / face scores, heatmaps, forensic modules
AudioForensicsView	LCNN score + per-window timeline as the main result. "Supporting signals (unvalidated)" below it
VideoForensicsView	Per-track face-swap scores, segment timeline, 11 stages with heuristics labelled unvalidated, no-face notice
*PipelineVisualizer	Stage-by-stage audit trail from `pipeline_modules`
AuthedImage	Loads evidence images with credentials

API client (src/lib/api.js)

- Every request uses `credentials: 'include'`. **No token in localStorage**; the httpOnly cookie carries the session.
- Report downloads and the WebSocket use the signed URLs from `POST /jobs/{id}/links` and request fresh ones when they expire.
- In development, Vite proxies `/api` (and WebSockets) to `localhost:8000`. In production, nginx does the same.

Testing

Vitest + Testing Library. Fixtures in `src/components/__tests__/fixtures/` are JSON exported from real analysis jobs (`real_audio_result.json`, `real_video_faceswap_result.json`, `real_video_faceswap_deepfake.json`, `real_video_noface_result.json`). Last run on this branch: 8 test files, 46 tests passed.

13 Security, privacy and deployment

Authentication & links

- JWT in an httpOnly cookie: `SameSite=Lax`, `Path=/api`, Secure in production.
- No `?token=` query parameters. Reports and the WebSocket use HMAC-signed URLs with `exp` and `sig`, scoped to one job, 5-minute TTL.
- Production boot fails with the placeholder or a short JWT secret (minimum 32 characters).

Abuse controls

- Upload rate limits per user and per IP (Compose defaults: 20/h users, 3/h guests).
- `X-Forwarded-For` is never trusted. `X-Real-IP` is trusted only from `TRUSTED_PROXIES`.
- Early 413 on `Content-Length`. Duration and media-type limits.
- Security headers middleware. HSTS in production.

Privacy & retention

- Stored: the upload, its SHA-256, evidence images, the result and reports.
- Guest jobs are deleted with all files after 72 h by the hourly `beat` purge.
- Registered users' jobs are kept until deleted. Deletion removes every derived file.

Error hygiene & observability

- Unhandled errors return an opaque message plus a `request_id`. The full traceback goes to server logs only.
- Validation errors never echo the raw body back.
- JSON logs in production, with a request-ID header on every response.

Deployment topologies

Mode	How	Notes
Local development (eager)	`uvicorn app.main:app` + `npm run dev`	SQLite, no broker. The API warms up models in GPU memory at startup. Frontend on :3000, API on :8000.
Full stack (queued)	`docker compose up --build`	Postgres, Redis, API, worker (concurrency 1, scene detectors preloaded), beat, nginx on :8080. Migrations run with `alembic upgrade head` before serving.
Image-only public beta	`deploy/docker-compose.prod.yml` + Caddy or Cloudflare Tunnel	`ENABLED_MEDIA_TYPES=image`, nightly `pg_dump` kept 7 days, disk alert at 80%, uptime check on `/api/health`. Steps in `deploy/README.md`.

14 Limitations and open items

Model-level

- **Audio** is evaluated only on ASVspoof 2019 LA (studio English read speech). It is weak on attack A17 (26.90% EER). Untested on phone, real-world or non-English audio.
- **Image** ensemble selection numbers live in a code comment, not a results file. The served ensemble is third-party; new generators may evade it.
- **Image face model** is excluded from verdicts (21% / 15% flag rates on real / AI faces). Its 99.98% validation accuracy is in-distribution only.
- **Video** verdict uses hand-set fusion weights with no labelled-video evaluation. Stages 1, 4–9 and 11 are unvalidated heuristics. Sampling at 1 fps (max 32 frames) can miss manipulated segments.
- **Face-swap model** is not on this branch. No metric is claimed.

Input-level (from docs/LIMITATIONS.md)

- Heavy JPEG compression, social-media re-encoding and screenshots destroy forensic signals.
- Low-bitrate (< 64 kbps), very short (< 2 s) or noisy audio reduces reliability. ENF needs audible mains hum.
- Metadata can be stripped or forged and is treated only as supporting evidence.
- SHA-256 proves a file hasn't changed since hashing. It does not prove the content is genuine.

Open engineering items

Item	Suggested resolution
Image ensemble numbers not in a results file	Run `ml/evaluation/eval_image_detectors.py`, commit its output, add rows to `paper_results/INDEX.md`.
Image model version mismatch (configured v3.0.0 vs checkpoint v2.1.0)	Align `IMAGE_MODEL_VERSION` with the checkpoint metadata, or replace the checkpoint.
Video fusion unvalidated	Merge `ml/faceswap-wire`, evaluate on the FF++ / Celeb-DF test splits, and let the trained model drive the video verdict.
Frontend lint debt	`npm run lint` reports pre-existing unused-import errors in files outside the video work.

A Appendix A: Provenance of every number

Per the project rules, every number above must trace to a file written by a script in the repository. Numbers marked COMMENT come from source-code comments only and should be regenerated into a results file before citing them.

Number(s)	File	Key / location
Audio EER 4.78%, AUC 0.9865, min t-DCF 0.1025, threshold 0.497, 7,355 / 63,882	`results/audio_eval.json`	`asvspoof2019_la_eval.*`, `model.decision_threshold_from_dev`
Audio 3-seed EER 5.26 ± 0.65%, AUC 0.9831, min t-DCF 0.1104 ± 0.0142	`paper_results/audio_asvspoof2019la/metrics.json`	`summary.*` (via INDEX.md)
Per-attack EER A07–A19 (incl. A17 26.90%)	`paper_results/audio_asvspoof2019la/metrics.json`	`summary.per_attack_eer_percent.*.mean`
Published baselines B01 9.57%, B02 8.09%	`paper_results/audio_asvspoof2019la/baselines.json`	`eval.B0x_*.eer_percent`
Dataset sizes (train 25,380 / dev 24,844 / eval 71,237)	`paper_results/audio_asvspoof2019la/dataset_summary.json`	`splits.*`
Heuristic dev AUCs, 0.6 inclusion rule, legacy verdict rates	`paper_results/audio_heuristics_dev/signal_auc.json`; `paper/config/frozen_audio_config.json`	`signals.*`, `heuristics_rule`
Audio checkpoint SHA-256 `7ca31fd0…`	`paper/config/audio_calibration.json`	`checkpoint_sha256`
Audio band 0.417 / 0.577	`docs/API_CHANGES.md` (ml/audio-wire)	derived from the frozen config
Face model validation metrics, 936,009 / 164,000, v2.1.0	`checkpoints/image_detector.summary.json`	`best_val_metrics.*`
Scene ensemble AUCs (0.973 / 0.984 / 0.836 / 0.469), 560 images COMMENT	`backend/app/ml/image_pipeline.py`	comment above `SCENE_DETECTORS`
Face model 21% / 15% flag rates COMMENT	`backend/app/config.py`	comment above `face_model_in_verdict`
Face-seam 9 of 30 (30%); CFA absent in 196 real JPEGs COMMENT	`backend/app/ml/image_pipeline.py`	section 5 comments
Thresholds 0.5 / 0.08, limits (100 MB, 600 s, 300 s, 72 h, 300 s TTL), 1 fps / 32 frames	`backend/app/config.py`, `docker-compose.yml`	`Settings` defaults
Frontend tests 8 files / 46 tests	`npm test -- --run` output, 24 Oct 2026	console summary

B Appendix B: Running the project

Local development (Windows, as used on this machine)

```
# Backend API (eager mode, SQLite, models warmed into GPU)
cd backend
..\.venv\Scripts\python.exe -m uvicorn app.main:app --host 127.0.0.1 --port 8000

# Frontend (Vite dev server with /api proxy)
cd frontend
npm run dev            # http://localhost:3000

# Tests
cd backend  && python -m pytest -q
cd frontend && npm test -- --run
cd frontend && npm run lint
```

Full stack with Docker

```
cp .env.example .env    # set JWT_SECRET_KEY (≥ 32 chars) and POSTGRES_PASSWORD
docker compose up --build
# Frontend  http://localhost:8080
# API docs  http://localhost:8000/docs
```

Key environment variables

Variable	Default	Effect
`CELERY_BROKER_URL`	empty	Empty: eager mode, API loads models. Set: queued mode, only the worker loads models.
`ENABLED_MEDIA_TYPES`	image,audio,video	Disabled types get 403.
`AUDIO_MODEL_SOURCE`	lcnn_asvspoof	`legacy` = hand-set rules, results marked experimental.
`FACE_MODEL_IN_VERDICT`	false	Keep false until a validated face model exists.
`FAKE_THRESHOLD` / `UNCERTAIN_BAND`	0.5 / 0.08	Image and video verdict band.
`VIDEO_SAMPLE_FPS` / `VIDEO_MAX_FRAMES`	1.0 / 32	Video sampling density.
`RETENTION_HOURS`	72	Guest job lifetime.
`SIGNED_URL_TTL_SECONDS`	300	Lifetime of report and WebSocket links.
`MAX_UPLOAD_MB`	100	Must match nginx `client_max_body_size`.

Disclaimer

DeepSentinel AI gives an automated technical assessment only. It is not forensic or legal evidence and makes no claim of compliance with any forensic or legal standard. Automated detectors make mistakes in both directions; treat every result as a lead, not as proof.